

Heart Failure

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Intended learning outcomes

- Understand the pathophysiology of heart failure including the compensatory mechanisms and the targets for treatment.
- Classify heart failure according to ejection fraction.
- Identify the causes of heart failure.
- Recognize the symptoms and signs of heart failure.
- Outline the investigations to be performed to diagnose heart failure.
- Outline the differential diagnosis of heart failure.
- Summarize the pharmacological treatment and device therapy of heart failure and explain their rationale and the practical considerations with their use.
- Recognize the acute heart failure presentation.
- Identify the causes and precipitating factors of acute heart failure.
- Outline the treatment of acute heart failure.

Heart failure

Definition

Heart failure (HF) is a **clinical syndrome** characterized by

- typical symptoms (e.g. breathlessness, ankle swelling and fatigue)
- signs (e.g. elevated jugular venous pressure, pulmonary crackles and peripheral oedema)
- caused by a structural and/or functional cardiac abnormality resulting in reduced cardiac output and/or elevated intracardiac pressures at rest or during stress.

Classification

- acute vs chronic
- right-sided vs left-sided
- high output heart failure
- According to symptom severity: NYHA functional classification:
 - Class I: No limitation of physical activity. Ordinary physical activity does not cause symptoms of HF.
 - Class II: Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in symptoms of HF.

- Class III: Marked limitation of physical activity. Comfortable at rest, but less than ordinary activity causes symptoms of HF.
- Class IV: Unable to perform any physical activity without symptoms of HF, or symptoms of HF at rest.
- According to ejection fraction (EF)
 - HF with reduced EF (HFrEF): $EF < 40\%$
 - HF with mildly reduced (HFmrEF): $EF\ 41\text{-}49\%$
 - HF with preserved EF (HFpEF): $EF \geq 50\%$

Etiology

- Diseases of the myocardium
 - IHD
 - inflammatory (post-myocarditis): infection (most commonly viral), immune-mediated
 - toxic: medications: antineoplastic drugs, recreational substance abuse: alcohol, cocaine, amphetamines
 - endocrinal: thyroid disease, adrenal disease, acromegaly, diabetes
 - nutritional: thiamine deficiency
 - electrolyte disturbance: hypocalcemia
 - infiltrative: amyloidosis
 - peripartum
 - stress-induced
 - genetic
- Abnormal loading conditions
 - HTN
 - valvular heart disease
 - congenital heart disease
 - high output states (severe anemia, sepsis, thyrotoxicosis)
 - volume overload (renal failure, iatrogenic)
- Pericardial disease
- Arrhythmias

Pathophysiology

Pathophysiology of HFrEF

- HFrEF is initiated after an index event reduces the pumping capacity of the heart. This results in reduced forward cardiac output as well as backward congestion.
- This triggers a series of compensatory mechanisms to maintain cardiac output. These are initially compensatory but eventually lead to a vicious circle of worsening heart failure. These are:
 - Neurohormonal activation: activation of the sympathetic nervous system and the renin-angiotensin-aldosterone system (RAAS). This leads to tachycardia, vasoconstriction and salt and water retention with increased blood volume.
 - left ventricular (LV) remodeling: LV hypertrophy with pressure overload, LV dilatation with volume overload as well as progressive myocyte loss and fibrosis

Pathophysiology of HFpEF

HFpEF is characterized by impaired LV relaxation and/or reduced compliance (increased stiffness) which results in elevated intracardiac pressures and backward congestion.

Clinical presentation

Symptoms

- dt left-sided congestion:
 - dyspnea: the severity of dyspnea from NYHA stage I to IV
 - paroxysmal nocturnal dyspnea
 - orthopnea
 - cough: dry or with white frothy, sometimes blood-tinged, sputum, more at night
- dt right-sided congestion: ankle swelling
- dt a reduced cardiac output: fatigue

Signs

- General examination:
 - tachycardia

- elevated jugular venous pressure
- bilateral pitting lower limb edema (sacral edema in bed-ridden patients)
- Cardiac examination:
 - S3 ± S4 gallop (most specific sign)
 - murmur
- Respiratory examination: bilateral (basal) pulmonary crepitations
- Abdominal examination: tender hepatomegaly

Investigations

- Laboratory tests:
 - complete blood count
 - serum sodium, serum potassium, serum urea, serum creatinine
 - liver function tests
 - fasting blood glucose, glycated hemoglobin
 - lipid profile
 - thyroid function tests
 - serum ferritin, iron, total iron binding capacity and transferrin saturation
 - Natriuretic peptides (NP): low levels can rule out the diagnosis of HF (a high negative predictive value). However, there are many cardiac and non-cardiac causes of elevated NP including atrial fibrillation, acute coronary syndrome, acute pulmonary embolism, advanced age, renal disease and liver disease. It should be noted that levels can be disproportionately lower in obese patients.
- ECG: to assess cardiac rhythm, rate, QRS morphology and duration and other abnormalities.
- Chest X-ray (CXR): can show cardiomegaly and pulmonary congestion as well as detect other potential causes of breathlessness.
- **Echocardiography**: is the key investigation to diagnose heart failure. It allows the assessment of the morphology and function of all cardiac

structures (myocardium, valves, pericardium), including the measurement of LVEF and estimation of LV filling pressures (diastolic function).

- Further investigations needed according to the etiology:
 - coronary artery disease: coronary computed tomography angiography, stress testing, invasive coronary angiography
 - cardiomyopathies and myocarditis: cardiac magnetic resonance imaging, specific investigations for etiology

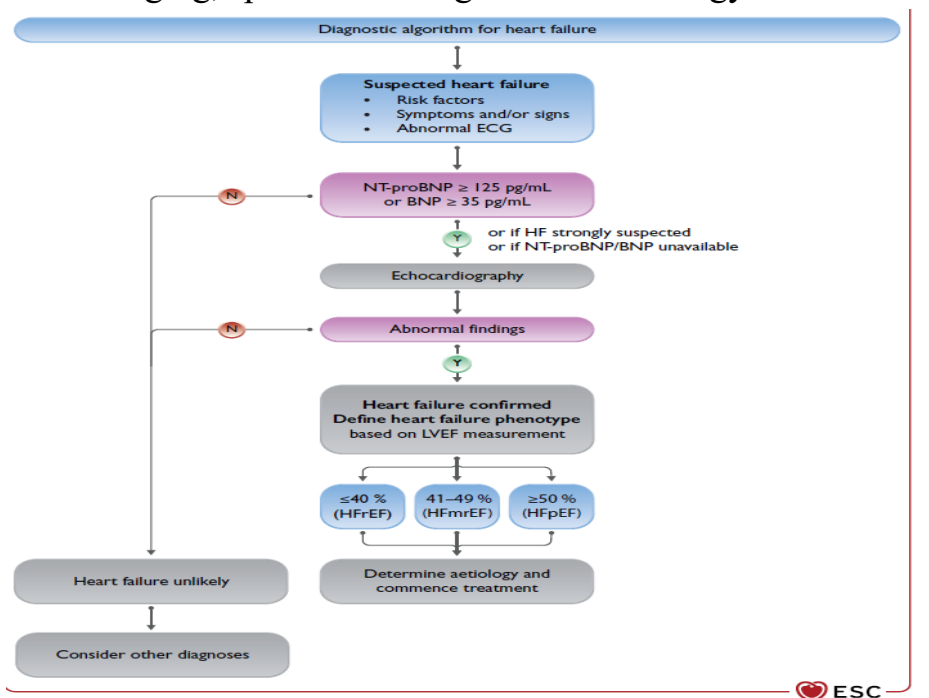


Fig. Diagnostic algorithm for heart failure. (From McDonagh et al. Eur Heart J. 2021.)

Differential diagnosis: respiratory diseases, others: anemia, thyroid disease, liver disease, renal disease, obesity, deconditioning

Treatment

Treatment of HFrEF

The aim of treatment is to:

- improve symptoms, functional capacity, quality of life
- reduce hospitalizations
- reduce mortality

- Non-pharmacological measures:
 - Patient education about the disease, the treatment prescribed and the importance of compliance as well as about drugs to be avoided (non-steroidal anti-inflammatory drugs (NSAID)).
 - Lifestyle advice includes diet (salt restriction), physical activity (adapted to symptom status) and smoking cessation.
 - Patients should be advised to monitor their weight twice weekly and be informed when to contact a healthcare professional.
 - Immunization against influenza (yearly) and pneumococcal disease.
- **Pharmacological treatment**
 - **Diuretics:** loop diuretics: to improve symptoms and signs of congestion
 - 4 classes of drugs should be prescribed in all patients to improve symptoms, reduce hospitalizations and reduce mortality:
 1. Angiotensin converting enzyme inhibitors
 2. Beta-blockers
 3. Mineralocorticoid receptor antagonists
 4. SGLT2 inhibitors
- 1. **Angiotensin-converting enzyme inhibitors (ACEI):**
 - ramipril, enalapril, captopril..
 - Mechanism: arterial vasodilatation → ↓ afterload, anti-RAAS, anti-LV remodeling
 - Contraindications: bilateral renal artery stenosis, history of angioedema, pregnancy
 - Used with caution in $GFR < 30 \text{ ml/min/1.73m}^2$, $Cr > 2.5\text{mg/dL}$, $K > 5.0 \text{ mmol/L}$, $SBP < 90\text{mmHg}$
 - titrated to maximum tolerated dose
 - follow-up: blood pressure, renal functions

Angiotensin receptor-neprilysin inhibitor (ARNI): sacubtril/valsartan: indicated as a replacement for ACEI if still symptomatic on ACEI. They may also be considered as first line instead of ACEI, especially in patients hospitalized with acute HF.

Angiotensin receptor blocker (ARB): indicated only if ACEI is not tolerated (ACEI-induced cough, history of angioedema and with caution in ACEI-induced angioedema)

2. Beta-blockers (BB):

- bisoprolol, carvedilol, metoprolol (succinate), (nebivolol)
- Mechanism: block sympathetic activity reducing heart rate and thus myocardial work
- Contraindications: heart block, critical limb ischemia, (asthma)
- initiated in clinically stable euvolemic patients, and slowly uptitrated
- follow-up: heart rate

3. Mineralocorticoid receptor antagonists (MRA):

- spironolactone, eplerenone
- Mechanism: diuresis, anti-RAAS, anti-LV remodeling
- follow-up: renal functions

4. SGLT2 inhibitors:

- dapagliflozin, empagliflozin
- mechanism: natriuresis, osmotic diuresis, + not fully elucidated mechanism

- Other drugs: ivabradine I_f channel inhibitor, digoxin and hydralazine and isosorbide dinitrate
- It should be noted that the following drugs can cause worsen HF and should not be used: NSAID and cox-2 inhibitors, corticosteroids, non-dihydropyridine CCB (verapamil, diltiazem), thiazolidinediones (glitazones).

- **Device therapy**

- Cardiac resynchronization therapy (CRT):
 - CRT improves cardiac function through the restoration of the synchrony of ventricular contraction by electrically activating the heart in a synchronized manner. Biventricular pacing is

achieved through the placement of three leads, in the right ventricle, the left ventricle (coronary sinus) and the right atrium.

- It is indicated in patients with HF who are symptomatic despite optimal medical therapy, in sinus rhythm, with LVEF $\leq 35\%$ and a wide QRS (QRS duration ≥ 130 ms and LBBB QRS morphology or ≥ 150 ms and non-LBBB QRS morphology).
- Implantable cardiac defibrillator (ICD) is indicated for secondary prevention of sudden cardiac death (SCD) in patients who have recovered from a ventricular arrhythmia causing hemodynamic instability and who are expected to survive for >1 year with good functional status, in the absence of reversible causes. It is also indicated for primary prevention of SCD in symptomatic HF (NYHA class II - III), especially of an ischemic etiology (> 40 days post-MI), and an LVEF $\leq 35\%$ despite ≥ 3 months of optimal medical therapy, provided they are expected to survive substantially longer than 1 year with good functional status.

- **Treatment of the cause, ppt factors and co-morbidities**

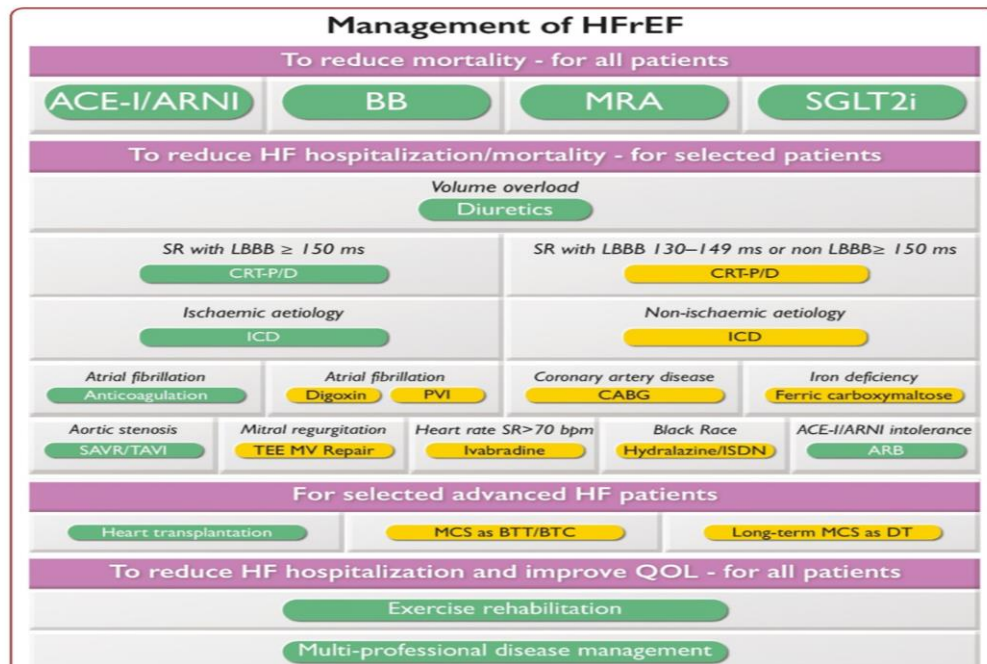


Figure. Management of HFrEF. (From McDonagh et al. Eur Heart J. 2021)

- Advanced heart failure is defined as severe symptoms and/or signs of HF at rest (NYHA IV), recurrent hospitalizations despite guideline-directed medical therapy, requiring advanced therapies as mechanical circulatory support, palliative care or cardiac transplantation.

Treatment of HFpEF

- Diuretics to relieve congestion
- SGLT2i: the only drug class shown to reduce mortality and morbidity in HFpEF
- Treatment of the cause, ppt factors and co-morbidities. These can include ACEI/ARB or MRA for hypertension and BB for atrial fibrillation or ischemic heart disease.

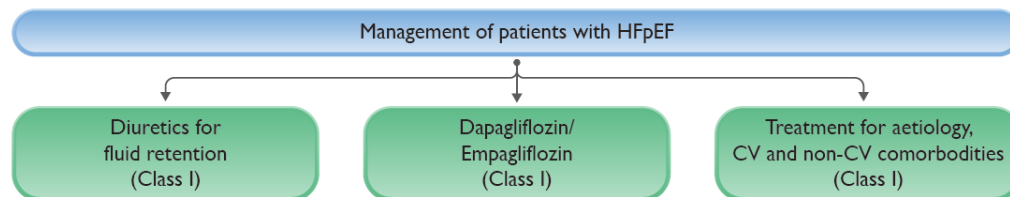


Figure. Management of HFpEF. (From McDonagh et al. Eur Heart J. 2023)

Acute heart failure

Acute HF is rapid onset or worsening of symptoms and/or signs of HF requiring urgent treatment and hospital admission. It can be de novo (new-onset) or an acute decompensation of chronic HF.

Etiology (causes/ppt factors)

Acute HF may be caused by primary cardiac dysfunction or be precipitated by extrinsic factors in patients with pre-existing cardiac dysfunction.

- Acute coronary syndrome
- Hypertension
- Arrhythmia: tachyarrhythmia or bradyarrhythmia
- Acute mechanical cause (myocardial rupture complicating ACS, acute native or prosthetic valve regurgitation secondary to endocarditis)
- Pulmonary embolism

- Infection (e.g. pneumonia, infective endocarditis, sepsis)
- Cardiac tamponade
- Drugs (e.g. NSAIDs, corticosteroids, negative inotropic substances, cardiotoxic chemotherapeutics)
- Non-adherence with salt/fluid intake or medications
- Toxic substances (alcohol, recreational drugs)
- Increased sympathetic drive, stress-related cardiomyopathy
- Metabolic/hormonal derangements (e.g. thyroid dysfunction, diabetic ketosis, adrenal dysfunction)
- Severe anemia
- Pregnancy and peripartum related abnormalities
- Surgery and perioperative complications

Clinical presentation

Symptoms

- acute dyspnea, orthopnea, PND, cough with frothy $\pm$ blood-tinged sputum

Signs

- General examination:
 - Vital signs: tachycardia, tachypnea $> 25/\text{min}$
 - elevated jugular venous pressure
 - bilateral lower limb edema
 - cold extremities, weak peripheral pulses in shock state
- Cardiac examination: S3 $\pm$ S4 gallop, murmur
- Pulmonary crepitations (extent: initially basal, may extend to lung apices)
- The initial examination should detect signs of respiratory or hemodynamic compromise (ABCDE approach to critically ill patient).

Clinical profiles

According to the peripheral perfusion state and congestion state, acute HF can present as one of 4 clinical profiles: warm & dry, warm & wet, cold & wet, cold & dry. Two severe forms are:

- Acute pulmonary edema: acute decompensated heart failure with a more rapid onset and severe pulmonary congestion

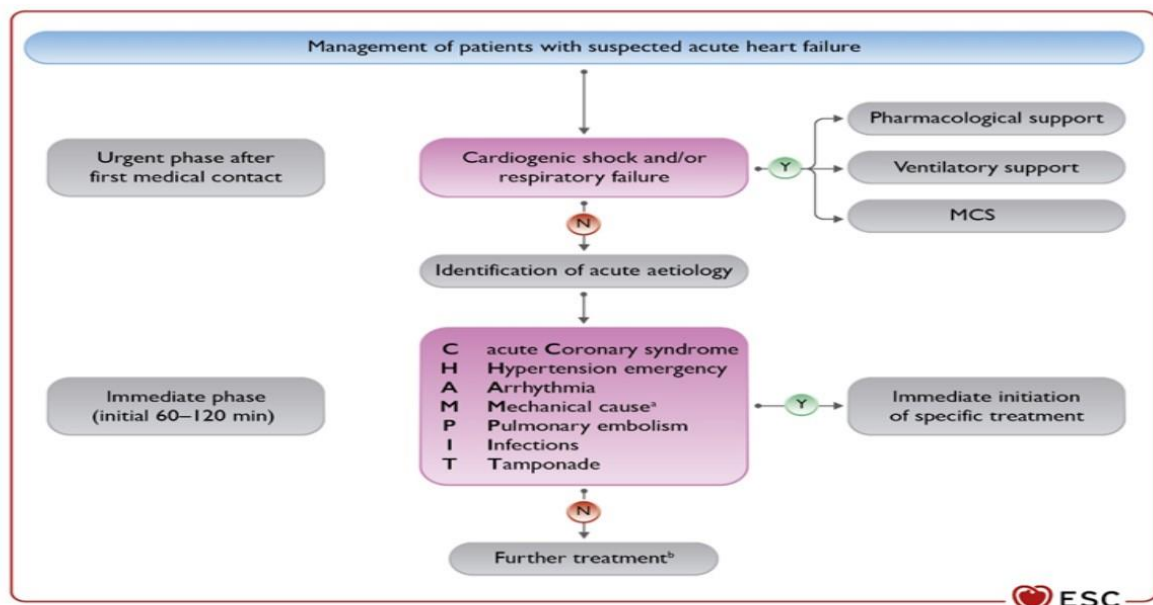
- Cardiogenic shock: with signs of hypoperfusion: cold extremities, weak peripheral pulses, mental confusion, oliguria

Investigations

- Laboratory tests: include routine tests, NP, troponin as well as arterial blood gas analysis in cases with respiratory distress or shock.
- ECG
- CXR
- Echocardiography

Treatment

- Oxygen and ventilatory support
 - Oxygen if $\text{SpO}_2 < 90\%$ or $\text{PaO}_2 < 60 \text{ mmHg}$
 - Non-invasive positive pressure ventilation
 - Intubation is recommended for progressive respiratory failure persisting despite oxygen administration or non-invasive ventilation
- Treatment of the cause/ppt factor
- According to the clinical profile (perfusion and congestion):
 - Congestion with adequate perfusion:
 - IV diuresis (loop diuretic): dose guided by the urine output
 - IV vasodilators (nitroglycerine) in acute pulmonary edema if $\text{SBP} > 110 \text{ mmHg}$
 - Cardiogenic shock (hypoperfusion):
 - vasopressors (norepinephrine) $\pm$ inotropes (dobutamine)
 - Short-term mechanical circulatory support: ex. intra-aortic balloon pump, extracorporeal membrane oxygenation, can be considered as a bridge therapy in case of no response to drugs.
- Thromboprophylaxis
- Pre-discharge and long-term: initiate and uptitrate evidence-based pharmacological therapy. A follow-up visit is recommended 1-2 weeks after discharge.



www.escardio.org/guidelines

2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure
(European Heart Journal 2021 – doi:10.1093/eurheartj/ehab368)

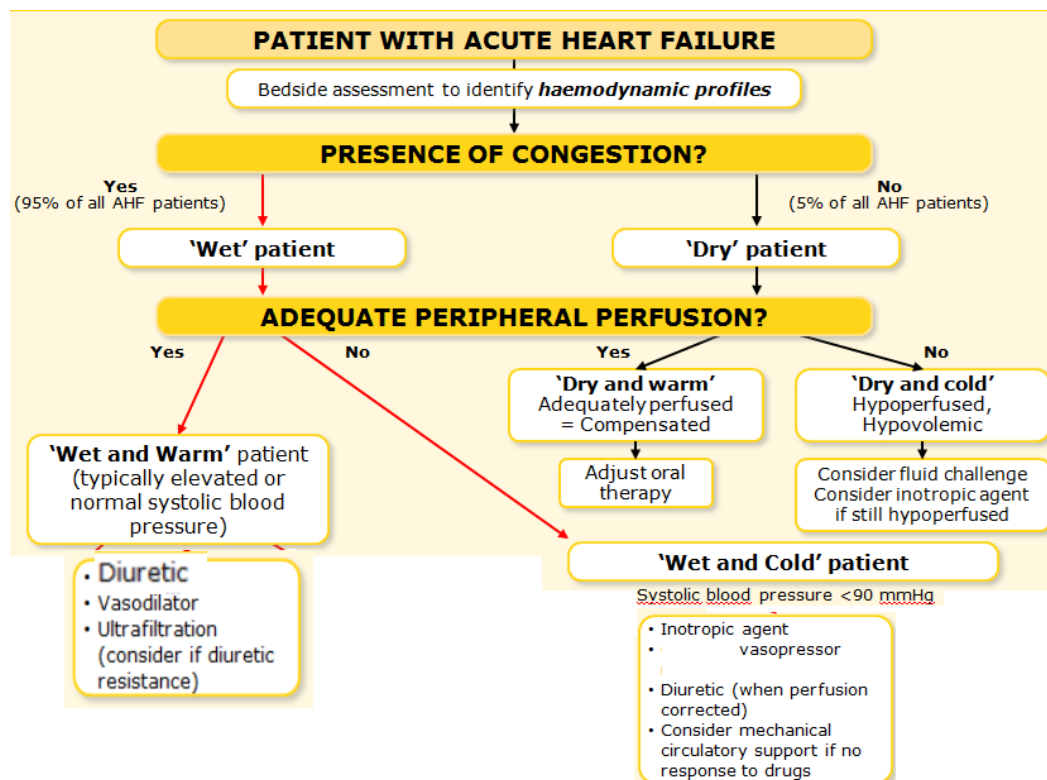


Figure. Management of acute heart failure. (From McDonagh et al. Eur Heart J. 2021 and Ponikowski et al. Eur Heart J. 2016)

References

McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure: Developed by the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) With the special contribution of the Heart Failure Association (HFA) of the ESC. *European Heart Journal*. 2021;42(36):3599-726.